

Cover Letter

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To: The Editor-in-Chief, *AEÜ — International Journal of Electronics and Communications*

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Manuscript title: Task-Oriented Single-Channel Blind Source Separation: Bit-Level Gains Are Interference-Limited, Not Loss-Limited

Manuscript type: Original Research Paper

Dear Editor,

We are pleased to submit the above manuscript for consideration in *AEÜ*. The paper is a *measurement study with an evaluation protocol contribution*: it asks, for single-channel blind source separation (SC-BSS) of co-frequency communication signals, whether the waveform-fidelity gains that the field optimises and reports (SI-SDR/SDR) actually reach the demodulated bits—and, if not, whether a task-oriented loss can make them. This question sits squarely within *AEÜ*'s readership: it connects the deep-learning separation literature to the classical co-frequency interference-cancellation and PCMA line of work, and it concerns exactly the physical-layer receivers your community builds. We ask you to consider it as a measurement/protocol paper whose negative results are *diagnosed*, not merely reported.

Key contributions.

1. **A reproducible empirical law along the interference axis.** On a variable-count ($K \in \{1, 2, 3\}$) co-frequency benchmark scored by an offset-compensated receiver (compensated SER and Gray-mapped BER), a +4.1 dB SI-SDR improvement moves SER only from 0.5049 to 0.5027— even +8.4 dB at −10 dB SNR buys nothing (if anything a fraction-of-a-point penalty)—because the mixture is interference-limited at $\text{SIR} \approx 0$ dB (one interferer raises the baseline SER $3.8\times$, from 0.146 to 0.561). A differentiable soft-demodulation loss whose receiver front-end is line-by-line identical to the evaluation receiver changes the pooled bit error rate by at most 0.03 pp in the fair additive comparison— but the pool hides a sign structure: the term *helps* at $K=2$ (+0.35 pp BER, 5/5 seeds positive in all four contrasts) and *hurts* at $K=3$ (−0.6 to −0.7 pp SER, 0/5 seeds positive), cancelling under the 1:2:3 pair weighting. The number of interferers, not the loss, sets the sign of what little effect exists—an experimental variable we are not aware of being isolated before in this literature.
2. **An evaluation protocol and statistical standard.** The training-time soft receiver and the evaluation receiver share one implementation, removing the train/eval mismatch that has made earlier SER claims hard to compare; all contrasts use an exact sign-flip permutation test at the seed level and a Monte-Carlo permutation test at the cell level, with Bonferroni control over eight pre-declared contrasts and the $n=5$ power ceiling ($p \geq 0.0625$ attainable) stated explicitly rather than smoothed over. We believe this protocol is reusable wherever separation is evaluated against downstream bits.
3. **A controlled attribution experiment and a concrete prescription.** The strongest task-oriented design—a joint soft-information head that bypasses explicit separation—fails

at the modulation-averaged chance level, and a controlled probe (same head, same data, oracle vs. blind synchronisation as the only variable) attributes the failure to burst-level blind carrier/phase recovery rather than to model capacity. The practical conclusion for the co-frequency community is direct: closing the waveform-to-bit gap requires explicit synchronisation and equalisation structure—the machinery of the interference-cancellation/PCMA tradition—not a better loss weight.

On reporting negative results. Several of our headline findings are nulls or adverse effects, and we report them as such, with mechanism. We submit them because the alternative—continuing to rank SC-BSS systems by SI-SDR alone—has a measurable cost: our per- K decomposition shows that pooled metrics can hide opposite-signed effects, and our audit of the standard recipe shows a configuration can be reward-hacked into waveforms that no receiver would accept. We would rather the field learn this from data than from another round of architecture papers optimising the wrong endpoint.

Reproducibility. All training, evaluation, analysis, and figure/table-generation code, including the synthetic data generator and the deterministic test grid (test seed 99999), is publicly available at https://github.com/BinNong/single_channel_source_seperation/tree/main/paper5_task_oriented; no external datasets are required. Every headline number is multi-seed (five seeds), and the exact command behind every number is recorded in the repository’s experiment log.

Related submissions. Two related companion manuscripts are under review elsewhere and share no results with the present paper: a closed-set separator design (the complex CNN with squeeze-and-excitation used as a baseline family here) at *Wireless Personal Communications* (Springer), and the open-world variable- K study that this paper’s benchmark and slot separator build upon at *China Communications*. The present manuscript is fully self-contained; the cited companion manuscript (currently listed as “under review”) is available to the editor and reviewers on request.

Originality and ethics.

- The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal.
- All authors have approved the submission; the authors have no conflicts of interest to declare.
- No human subjects or animal experiments are involved; all experiments use synthetic signals from our released generator.

Author contributions. B. Nong conceived the measurement study and the task-oriented formulation, ran the multi-seed experiments and statistical analyses, and drafted the manuscript. W. Fu contributed to the conceptualisation of the evaluation framework. Z. Jiang contributed resources and validation. All authors reviewed and approved the final manuscript.

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Suggested reviewers. We have no specific reviewer suggestions; we trust the editorial board to identify expert reviewers in co-frequency interference management and machine-learning-based physical-layer signal processing.

Thank you for considering our manuscript. We look forward to hearing from you.

Sincerely,

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